

Juana Granados Rodríguez

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GitHub

Profile

I am a recent physics graduate with a strong interest in theoretical and mathematical physics. My undergraduate thesis explored the mathematical foundations of wavefront sets, reflecting my passion for abstract structures and rigorous formalism. In parallel, I have gained hands-on experimental experience in nanomagnetism, contributing to an ongoing publication. This blend of theory and lab work allows me to approach physical phenomena from both foundational and applied perspectives. I value interdisciplinary collaboration, precision in analysis, and continuous intellectual challenge.

Education and Training

Universidad de los Andes

Bogotá, Colombia

Bachelor's Degree in Physics (GPA: 4.28/5.00)

2020 - 2024

- Relevant coursework: Electromagnetism, Quantum Mechanics, Renormalization, Statistical Physics, Computational Methods, Particle Physics.
- Recipient of the *Préstamo Condonable para Carreras Especiales* by the Universidad de los Andes and fully condoned due to outstanding academic performance.

Minor in Mathematics

- Relevant coursework: Discrete Mathematics, Advanced Linear Algebra, Abstract Algebra, Complex Variable Calculus, Probability.

Projects and Research Experience

QFT and Mathematical Physics Group and Seminar

Universidad de los Andes

Active Participation in Seminar

September 2023 - July 2024

- Participated in weekly discussions on advanced topics in quantum field theory.
- Shared advancements in my learning and collaborated with peers and professors to deepen my understanding of quantum field theory.

Undergraduate Thesis

January 2024 - August 2024

- Analyzed deeply the mathematical structure of wavefront sets in quantum field theory and applied Epstein-Glaser techniques to explore renormalization processes in theoretical physics under the supervision of Professor Andres Fernando Reyes Lega, Ph.D, Associate Professor at Universidad de los Andes. [View Document](#)

Nanomagnetism Group and Seminar

Universidad de los Andes

Research Assistant

June 2024 - Present

- Presented results during seminar and group meetings.
- Studying Co-doped V_2O_5 nanoparticles using VSM, XRD, and Raman Spectroscopy.
- Performing experimental characterization and contributing to the preparation and drafting of a scientific paper for publication in a peer-reviewed journal.

Experimental Project

September 2023 - December 2023

- Investigated electric and magnetic properties of Co-doped V_2O_5 under the supervision of Professor Juan Gabriel Ramirez Rojas, Ph.D., Associate Professor at Universidad de los Andes.
- Participated in weekly seminars and discussions within the research group.
- Conducted experiments using Vibrating Sample Magnetometry (VSM), X-ray Diffraction (XRD), and Raman Spectroscopy for material characterization.
- Analyzed experimental data and contributed to interpreting of results in the context of nanomagnetism.

Work Experience

Teaching Assistant - Honors Integral Calculus and Differential Equations

Universidad de los Andes, Bogotá, Colombia

August 2021 - December 2021

- Graded assignments and exams, assisted the professor in virtual classes and developing class materials and evaluations, addressed students' questions online, and managed the course page and resources.

Teaching Assistant - Vector Calculus

Universidad de los Andes, Bogotá, Colombia

June 2023 - August 2023

- Graded assignments and exams, held regular office hours to guide students through challenging concepts, and provided academic support to ensure their understanding and progress.

Skills and Interests

Programming and Technical Tools: Python (NumPy, SciPy, Matplotlib, Pandas, Scikit-Learn), $\text{\LaTeX}$, R, Git, GitHub, Jupyter Notebooks

Data Analysis and Tools: Computational modeling, data visualization, statistical analysis, Python for data analysis, experience with VSM, XRD, and Raman spectroscopy

Experimental Skills: Familiar with laboratory protocols, vibrating Sample Magnetometry (VSM), X-ray Diffraction (XRD), Raman Spectroscopy, and nanoparticle synthesis techniques.

Languages: Spanish (Native), English (IELTS Academic 7.5 band score)

References

Andres Reyes, Ph.D

Associate Professor, Universidad de los Andes

Email: anreyes@uniandes.edu.co

Juan Gabriel Ramirez, Ph.D

Associate Professor, Universidad de los Andes

Senior researcher, MinCiencias

Member Materials Research Society

Member American Physical Society

Advisory Editorial Board member of the Journal of Magnetism and Magnetic Materials

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